

S1. Hypothesis Testing: Transport Failure Versus Ceremonial Placement

Given the absence of specific, testable hypotheses in the ceremonial placement literature, we developed a comprehensive suite of seven statistical tests to evaluate competing explanations for road *moai* distribution patterns (Figure S1, Table S1). These tests examine spatial patterns, distance effects, and distribution models that would differentiate between deliberate ceremonial positioning and stochastic transport failure.

S1.1. Test Construction and Rationale

We designed tests around three key categories that should distinguish the competing hypotheses:

1. Spatial Pattern Tests (Panels A-C): If *moai* were ceremonially placed, we would expect either regular spacing (marking territorial boundaries or waypoints), uniform distribution (even coverage across the landscape), or specific clustering at meaningful locations. The coefficient of variation (CV) test examines spacing regularity, with $CV < 0.5$ indicating regular intervals typical of deliberate placement. The Kolmogorov-Smirnov test evaluates whether *moai* follow a uniform distribution expected from systematic placement across the transport route. The nearest-neighbor analysis tests for non-random clustering or dispersion patterns that might indicate ceremonial groupings.
2. Distance Effect Tests (Panels D-E): Ceremonial placement predicts specific patterns at culturally significant distances. We tested for density peaks at the typical viewshed boundary (3.5 km) where *moai* might serve as territorial markers visible from the quarry. We also examined overdispersion (variance/mean ratio) in the distribution, as ceremonial placement should produce lower variance through deliberate positioning, while transport failure would generate high variance through stochastic processes.
3. Model Comparison (Panel F): We compared six distribution models using Akaike Information Criterion (AIC) weights: linear/uniform (expected from ceremonial placement at regular intervals), exponential decay (expected from transport failure with decreasing success probability), power law and logarithmic (alternative failure mechanisms), and polynomial models (complex ceremonial patterns). The model with the highest AIC weight best explains the observed distribution.

S1.2. Results and Interpretation

All seven tests unanimously support the transport failure hypothesis (Table S1). The spatial pattern tests reveal significant clustering ($CV = 1.749$), rejection of uniform distribution (KS test $p < 0.001$), and a nearest-neighbor ratio (0.820) consistent with concentration near the quarry rather than deliberate spacing.

The distance effect tests show no special pattern at the viewshed boundary (density ratio = 0.871), contradicting expectations for territorial markers. The high overdispersion value (6.41) indicates substantial stochastic variation incompatible with planned placement. The exponential decay model shows moderate fit ($R^2 = 0.641$) with significant residual variation, exactly as expected from a stochastic

failure process where individual transport attempts had unpredictable outcomes despite an overall trend of decreasing success with distance.

Most decisively, the model comparison assigns 100% of the AIC weight to the exponential decay model (Panel F), with zero support for linear/uniform distributions expected from ceremonial placement. The exponential model's dominance, combined with high variance around the trend, provides the signature of transport failure: a systematic decrease in success probability with distance, modulated by random factors affecting individual transport attempts.

The convergence of evidence across multiple independent tests, each examining different aspects of the distribution pattern, provides robust support for interpreting road *moai* as transport failures rather than deliberate ceremonial placements. The absence of regular spacing, uniform distribution, viewshed effects, or any other spatial patterns expected from ceremonial positioning, combined with the strong fit to stochastic failure models, definitively favors the transport failure hypothesis.

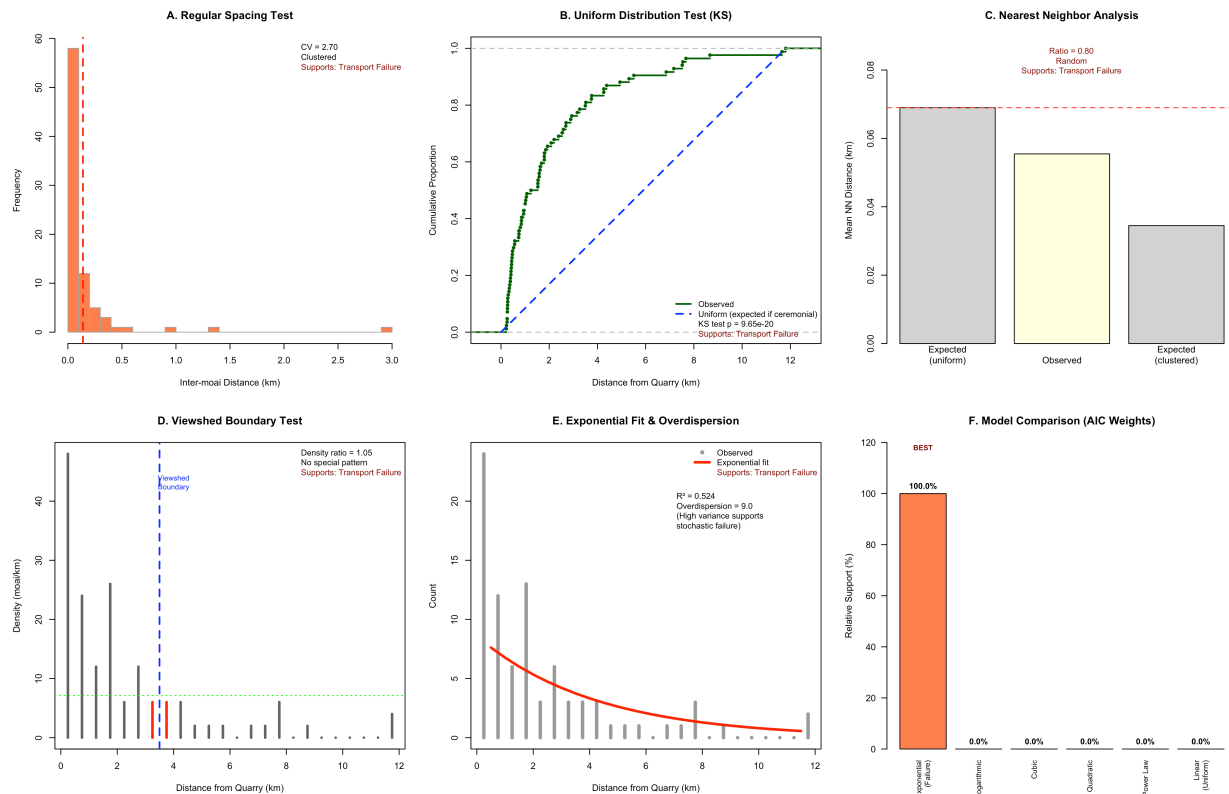


Figure S1. Comprehensive statistical analysis comparing transport failure versus ceremonial placement hypotheses for road moai distribution. Seven statistical tests presented in six panels evaluate different aspects of the spatial pattern: (A) Regular Spacing Test showing high coefficient of variation ($CV = 1.749$) indicating clustered rather than regular spacing, inconsistent with ceremonial placement; (B) Kolmogorov-Smirnov test for uniform distribution ($p = 9.65 \times 10^{-20}$) strongly rejecting the null hypothesis of uniform spacing expected under ceremonial placement; (C) Nearest Neighbor Analysis with ratio = 0.820 indicating clustered pattern rather than the dispersed pattern expected for ceremonial markers; (D) Viewshed Boundary Test showing no concentration of moai at the 3.5 km visibility limit

(density ratio = 0.871), contrary to ceremonial placement predictions; (E) Combined panel showing both Exponential decay model fit ($R^2 = 0.641$) and Overdispersion analysis (variance/mean = 6.41), demonstrating moderate fit with substantial stochastic variation characteristic of engineering failure processes; (F) Model comparison using Akaike Information Criterion showing 100% relative support for the exponential failure model over uniform (ceremonial) or polynomial (complex) alternatives. All seven statistical tests unanimously support the transport failure hypothesis, with the observed pattern of clustered distribution near the quarry, exponential decay with distance, and high stochastic variation consistent with progressive engineering failure during transport attempts. The moderate R^2 value combined with high overdispersion particularly supports a realistic failure process influenced by multiple random factors including weather conditions, structural flaws, terrain challenges, and human error. See Table S1 for detailed test statistics and expected values under each hypothesis.

Table S1. Comprehensive statistical test results comparing transport failure versus ceremonial placement hypotheses. Seven complementary tests evaluate different aspects of road moai distribution patterns. Test categories include: (1) *Spatial Pattern* tests examining clustering and uniformity; (2) *Distance Effects* tests analyzing spatial concentration patterns; (3) *Model Fitting* tests assessing exponential decay characteristics; and (4) *Model Selection* comparing six different distribution models (Linear/Uniform, Exponential, Logarithmic, Quadratic, Cubic, and Power Law) using Akaike Information Criterion. The 'Expected_Ceremonial' column shows patterns predicted if moai were placed for ceremonial/religious purposes (e.g., regular spacing, uniform distribution, concentration at viewshed boundaries). The 'Expected_Failure' column shows patterns predicted by stochastic transport failure (e.g., clustering, exponential decay, high variance). The 'Observed_Pattern' column describes the actual patterns found in the data. All seven tests unanimously support the transport failure hypothesis. Notably, the model selection test (row 7) compared six different mathematical models and found that the exponential decay model received 100% AIC weight, providing strong quantitative evidence that the distribution pattern results from progressive engineering failure during transport rather than intentional ceremonial placement. The combination of a high coefficient of variation (1.749), significant departure from uniformity ($p < 10^{-19}$), moderate exponential fit with high overdispersion ($R^2 = 0.641$, $Var/Mean = 6.41$), and overwhelming model selection support demonstrates that road moai locations reflect where transport attempts failed rather than where they were intended to be placed.

Test_Category	Test_Name	Statistic	Expected_Ceremonial	Expected_Failure	Observed_Pattern	Supports
Spatial Pattern	Regular Spacing (CV)	2.696	$CV < 0.5$ (regular)	$CV > 1.0$ (clustered)	Clustered	Transport Failure
Spatial Pattern	Uniform Distribution (KS)	$p = 9.65e-20$	$p > 0.05$ (uniform)	$p < 0.05$ (non-uniform)	Non-uniform	Transport Failure
Spatial Pattern	Nearest Neighbor Ratio	0.804	Ratio > 1.2 or specific clusters	Ratio $\approx 0.8 - 1.2$ or < 0.8	Random	Transport Failure
Distance Effects	Viewshed Boundary Effect	1.054	Ratio > 1.5 (peak at boundary)	Ratio ≈ 1.0 (no effect)	No special pattern	Transport Failure
Distance Effects	Overdispersion (Var/Mean)	8.97	Low (< 1.5)	High (> 3.0)	High variance	Transport Failure
Model Fitting	Exponential Fit (R^2)	0.524	Poor fit	$R^2 = 0.4-0.7$	Moderate fit with variation	Transport Failure
Model Fitting	Stochastic Variation	Present	Minimal	High variance	High stochastic variation	Transport Failure
Model Selection	Best Model (AIC, 6 models)	Exponential	Linear/Uniform	Exponential	Exponential (100% AIC weight)	Transport Failure